

DC Motor: Torque, Current and Rotation

DC Motor · oPhysics

Senior Secondary (Years 11-12)

~30 minutes

Science · Physics

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Simulation: <https://ophysics.com/em10.html>

Curriculum links: electromagnetism · magnetic fields

Learning intentions

- I can explain why a current-carrying coil in a magnetic field experiences a torque.
- I can describe the role of the commutator in keeping the coil rotating in one direction.

Getting started

Open the DC Motor simulation. Set the voltage, magnetic field strength and number of coil loops, then run the motor and watch it rotate.

Tasks

1. Run the motor at a low voltage, then increase the voltage. Describe the effect on the motor's rotation speed.
2. Reset the voltage, then increase the magnetic field strength instead. Describe the effect on the motor's speed and compare it to Task 1.
3. Reset again, then increase the number of loops in the coil. Describe the effect on the torque and speed.
4. Based on what you have observed, explain what would happen to the coil's rotation if the commutator were removed.

Extension (optional)

The torque on the coil is given by $\tau = N \cdot I \cdot A \cdot B \cdot \sin\theta$. Use this equation to explain your observations from Tasks 1–3.

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